EVALUATION OF SUPERCRITICAL CARBON DIOXIDE SPRAY TECHNOLOGY TO REDUCE SOLVENTS IN A WOOD FINISHING PROCESS

This paper has been reviewed in accordance with the U.S. Environmental Protection Agency's peer and administrative review policies and approved for presentation and publication.

Paul M. Randall
U.S. Environmental Protection Agency
Risk Reduction Engineering Laboratory
26 West Martin Luther King Drive
Cincinnati, OH 45268

INTRODUCTION

This project was conducted by the Pollution Prevention Research Branch(PPRB) of the U.S. Environmental Protection Agency(EPA) with the cooperation of Union Carbide Corporation, Nordson Corporation , and the Pennsylvania House Furniture Company. The PPRB is evaluating and demonstrating new technologies for pollution prevention through the Pollution Prevention Clean Technology Demonstration (CTD) Program. (1)

This paper reviews the use of supercritical CO_2 technology for paint spray application. Penhsylvania House Furniture Co. has used supercritical CO_2 coating technology to apply a nitrocellulose lacquer finish to oak and cherry furniture on a chair-finishing line. At current Pennsylvania House production rates, more than 250 furniture units per day are coated with nitrocellulose lacquer by this process.

During the evaluation, three issues of this technology were examined:

- Product Quality: to show that coating applied by this spray technology meets company standards for a quality finish.
- Pollution Prevention Potential: to demonstrate that use of this spray application technology for solvent replacement in coatings reduces volatile organic compounds (VOCs) released during finishing operations.
- Economics: to document the cost to install and operate this technology on an existing spray coating finish line.

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PROCESS DESCRIPTION

Supercritical fluids are gases that exist at temperatures and pressures near or above the critical point of the fluid as depicted on a phase diagram (Figure 1). At the critical point, the properties of the liquid and the gas are similar or identical. The resulting single-phase fluid exhibits solvent-like properties that can be altered by adjusting temperature and pressure. A number of gases have been examined for use as supercritical fluids in applications such as industrial and analytical separation

processes, cleaning, chromatography, and coating. The UNICARB™ process for coating uses nontoxic, nonflammable carbon dioxide as the supercritical fluid for coating dilution. Carbon dioxide, readily available as a by-product of a variety of industrial processes, has a critical temperature of 31.3°C (88°F) and a critical pressure of 72.9 atm (1070 psi), falling within the ranges already used for heated paint systems and airless spray equipment.

Union Carbide Corporation developed the use of supercritical CO_2 for spray coating applications, introducing this technology commercially in 1988 under the UNICARB™ tradename.(2,3) In the UNICARB™ process, the solvent-like properties of supercritical CO_2 are exploited to replace a portion of the organic solvent in the conventional solvent-borne coating formulation. The supercritical CO_2 acts as a diluent solvent to thin the viscous coating just prior to application, so that the coating can be atomized and applied with a modified spray gun. According to Union Carbide, 30 to 80% of the organic solvent in a coating formulation can be replaced with supercritical fluid. Typically, most of the volatile, fast-drying solvents and some of the medium-drying solvents are eliminated, retaining enough medium- and slow-evaporating solvents to obtain proper leveling and film coalescence. The solvent blends may need to be adjusted to optimize performance with the supercritical CO_2 spray technology. This usually can be done without changing the resin chemistry or pigment-loading levels. The solvent level of even conventional high-solids coatings can be reduced further when applied by this process. The actual reduction in solvent content that can be achieved is dictated by a number of factors. These include the type of coating being applied and its exact formulation, the desired film thickness and properties of the applied coating, and the environment in which the coating is being applied.

Thermosetting, thermoplastic, air-dry, and two-component formulations, in clear, pigmented, and metallic coating systems, have been developed successfully for application by supercritical CO_2 spray technology. Limitations exist with pigmented systems because some of the pigments, (e.g., carbon black), may be soluble in the supercritical CO_2. However, other pigments, including aluminum flake, titanium dioxide, and calcium carbonate, have been included successfully in formulations applied using this process. Nitrocellulose, silicone alkyds, acrylics, and a two-part urethane formulation have been developed for supercritical CO_2 application. Union Carbide is currently working on a two-part epoxy system and a phenolic resin formulation.

The supercritical temperature and pressure of CO_2 are within the ranges already used for heated paint systems and airless spray equipment, but special equipment is needed to introduce the CO_2 into the reduced solvent formulations and then heat and pressurize the resultant mixture prior to spraying. Typically, 10 to 50% carbon dioxide by weight may be introduced depending on the solubility in the coating, the solids level, the pigment loading, and the ambient conditions in the spray booth. Heating lowers the coating viscosity for easier pumping but decreases the solubility of the CO_2 in the coating concentrate. Therefore, optimum operating temperature and pressure must be determined and maintained to achieve the best results in each application.

Usually, the coating is heated to 40° to 70°C with spray pressures of 1200 to 1600 psi. The specially formulated coatings are applied with spray guns similar to those used for airless applications. However, because the decompression of supercritical CO_2 results in finer atomization of the sprayed coating and smaller particles than is common with use of airless spray equipment, slight modification of the spray gun nozzle design was required to optimize the spray pattern. The result is a more finely dispersed and more uniform pattern than typically is achieved with conventional air spray equipment.

Nordson Corporation designed the spray guns used by Pennsylvania House. A supply unit mixes coating concentrate and carbon dioxide to a desired ratio, pressure, and temperature, and delivers the solution to specially designed spray guns. The size of the supply unit is dictated by production requirements. A microprocessor-based controller continuously monitors the system and allows the operator to adjust the ratio of coating concentrate to carbon dioxide for best results. The

supply unit and controller are placed adjacent to, but outside of, the spray booth. The system equipment is available for either manual or automatic operation. Manual and automatic spray guns for electrostatic and non-electrostatic applications can be used with minor modifications to the nozzle. Because the reduced solvent coating is more viscous, a special pumping station is required.

Pennsylvania House uses the UNICARB™ process on the chair-finishing line at its plant in White Deer, Pennsylvania, to apply nitrocellulose lacquer finishes. This supercritical CO_2 spray technology has allowed Pennsylvania House to continue using the solvent-borne nitrocellulose lacquer coating that is used widely in the U.S. wood-finishing industry, while reducing VOC emissions from their finishing operation. To bring this technology to production-line use, Pennsylvania House worked closely with Union Carbide to optimize the basic process, Nordson for equipment-related issues, and with Guardsman and Lilly to optimize the formulations of reduced solvent coatings.

The chair-finishing line at the White Deer facility carries chairs, stools, and mirrors from assembly through the finishing process and to packaging. The finishing process is labor intensive, with manned stations for staining, wiping, rubbing, sanding, polishing, and inspection. The overhead conveyor system runs through the various work stations at 6 to 7 ft per minute; 6 ft per minute is approximately 60-70% of capacity. At this speed, 250-300 units per day are produced. Total time on the line from start to finish is about 4 hours.

Two color stains usually are used to highlight the natural grain and provide color to the wood. Toner stain is sprayed on first, followed by a sprayed mineral-spirit wiping stain, which then is wiped off by hand. Some pieces get a spatter stain for special effects before entering the oven for the first drying step. Oven temperature is maintained at 110°F for all heating steps.The next step is spray application of a 20%-solids nitrocellulose sealer followed by a second pass through the oven. Light hand sanding is performed if needed before or after application of the sealer.

The nitrocellulose lacquer is spray applied next. In the conventional finishing process, nitrocellulose lacquer (21-23%-solids) is applied manually using airless spray equipment in two coats, with a pass through the oven between the coats. Flash-off time between spraying the coating and entering the oven is 10 to 12 minutes. Oven residence time is 7 minutes. The same flash-off and oven-time intervals are used after the second coat of lacquer. Furniture units remain on the line at ambient temperatures for one hour and forty-five minutes after the last oven pass before they are packaged. When the supercritical CO_2 finishing process is used, only one coat of nitrocellulose lacquer is needed to achieve the desired film build and finish quality. The nitrocellulose lacquer formulation optimized for the Pennsylvania House production line has approximately a 41%-solids content. This coating is applied using the UNICARB™ equipment in spray booth #1, followed by a 10- to 12-minute flash-off period, and a 7-minute pass through the oven. Because Pennsylvania House has not reconfigured the chair conveyor line, the UNICARB™ line follows the existing conveyor line through the unused spray booth #2 and the final stage of the oven before reaching inspection and packaging for shipment. The second oven pass is not required for this process but has no negative effect on the cured finish coat.

Conventional and reformulated nitrocellulose lacquer coatings used by Pennsylvania House are supplied to the spray guns directly from the shipping drums. The drums of coating are equilibrated and mixed in the Pennsylvania House paint room before being pumped through lines to the spray booths. Temperature and humidity changes in the plant sometimes require adjustments in the solvent blend of the formulations for optimal spray results. With conventional lacquer, solvent is added to the drum of coating, which then takes about 1 hour to reach the spray booth. This adjustment process sometimes must be repeated to get the desired results. Once extra solvent has been added to a drum of coating, the contents of the drum have a higher volatile content. With the supercritical CO_2 spray system, the

operator simply adjusts the ratio of CO_2 to coating concentrate at the control unit located just outside the spray booth. No additional solvents are introduced into the process, and adjustments are immediately evident.

PRODUCT QUALITY EVALUATION

The specific objective of the product quality evaluation was to determine whether nitrocellulose lacquer applied by the supercritical CO_2 process provides a wood finish of equal or better quality than does the conventional nitrocellulose formulation and spray technique previously used by Pennsylvania House Furniture Company. At Pennsylvania House, the appearance and quality of the final finish are judged through visual examination by inspectors on the coating line. Special attention is given to gloss, smoothness, and the lack of surface defects such as blisters or pinholes.

Product quality was evaluated through subjective evaluations performed by Pennsylvania House staff members and a panel of Battelle coatings personnel. Test substrates were finished on the production lines during an on-site visit. A set of samples was finished using the one-coat UNICARB™ process, and two other sets were finished using one and two coats of the "standard" nitrocellulose formulation and the airless spray equipment still in place on the chair-finishing line. All panels were finished by the same production methods that typically are used on the chair line at Pennsylvania House.

The subjective product quality evaluations demonstrated that a coating applied by the supercritical CO_2 spray process yielded a product with a finish quality equal to or better than the finish quality obtained by conventional materials and methods. Samples finished by the supercritical CO_2 process and by the two-coat conventional process were rated as "acceptable." Samples sprayed with one coat of the nitrocellulose by the conventional process were not acceptable.

In addition to the subjective visual inspections of the test samples, Battelle staff made measurements of gloss(Table 1) using ASTM D529. These measurements provide some quantitative insight into the physical attributes of the finish line of each of the coating processes. The test procedure outlined in ASTM D529 recommends averaging six gloss measurements for a 3-inch x 6-inch sample area, which correlates to 49 measurements on the 7-inch x 21 -inch test substance used here. The mean and standard deviation of the 49 data points represent the overall gloss appearance of each sample, alleviating subjective biases of the person performing the measurements while still incorporating the assessment of any nonuniformity in the gloss across the sample surface. The breadth in standard deviation of the data can be used as a gauge of the uniformity of the sample finish across the complex geometry of the test panels. Gloss test results for each of the nine panels are reported as the mean of 49 determinations and then averaged for each of the sample sets for easy comparison among each of the finishing processes in Table 1. The averaged gloss data for the UNICARB™ samples are statistically the same as those for the conventional two-coat process. The gloss data of both of these sets show that they are substantially glossier than the one-coat conventional finish sample set.

These results are supported by Pennsylvania House records for consumer acceptance. Internal quality control audits on chair-line products show a decrease in finish defects using the supercritical CO_2 system. Based on the number of furniture units requiring rework because of finish defects, product efficiency has improved since the UNICARB™ process was implemented.

POLLUTION PREVENTION POTENTIAL

The pollution prevention potential of this technology is based on reducing the emissions of organic solvents without adding to other wastestreams. The nitrocellulose lacquer finishing process used on the chair line can contribute to pollution in two ways: VOC emissions from the coating formulation, and spray booth wastes, including solvent-laden filters and nitrocellulose "dust". In conventional spray coatings, a blend of fast-evaporating solvents, medium-evaporating solvents, and slow-evaporating solvents are used. In the supercritical CO_2 spray process, most of the fast- and medium-drying solvents are replaced by supercritical CO_2 and the slow-drying solvents are adjusted slightly for better film formation. Although reducing VOC emissions is important, it is equally important to demonstrate the supercritical CO_2 process does not add pollutants to other wastestreams.

Pennsylvania House has been able to reduce the number of coats of nitrocellulose lacquer from two to one. The volume of nitrocellulose lacquer used in each finishing operation was determined during the initial phases of implementing the supercritical CO_2 process at Pennsylvania House. Metering devices were placed in-line on the airless spray guns used to apply the conventional nitrocellulose formulation, and on the coating inlet line to the supercritical fluid supply unit used to feed the coating mixture concentrate and the supercritical CO_2 to the modified spray guns used with the UNICARB™ process.

Pennsylvania House records indicate that it takes approximately 473 ml of the conventional formulation to apply the two coats needed to achieve the desired quality in the finished product. The UNICARB™ process required about 207 ml of the reduced solvent formulation per furniture unit to achieve the same quality. There are two reasons that a smaller volume of coating is required when the supercritical CO_2 process is used: 1. the higher solids content of the UNICARB™ formulation means that more resin is transferred to the substrate per volume of formulation sprayed and 2. the increased viscosity of the film deposited by the process inhibits film buildup by soaking into the wood substrate.

Table 2 compares the volatile solvent content(% by weight) of the two formulations used by Pennsylvania House. The UNICARB™ coating is formulated using 17.5% less solvents (on an absolute basis) than the conventional formulation. Only 9.67% of the UNICARB™ formulation is comprised of Hazardous Air Pollutant (HAP) materials compared to 35.78% for the conventional formulation. On a per-gallon-of-coating-sprayed basis, this difference would result in a relative decrease in VOC emissions of 22.81%, with a 72.97% decrease in HAPs using the UNICARB™ formulation. VOC contents are reported as 563 gms/l for the UNICARB™ formulation and 707 gms/l for the conventional system.

Assuming an average yearly production of 50,000 units and the use of 207 ml for the one-coat UNICARB™ process and 474 ml for the two-coat conventional formulation, the UNICARB™ formulation uses 10,220 liters and the conventional formulation uses 24,604 liters to finish the units. Based on the reported VOC contents, this system change corresponds to an annual reduction in VOC emissions of 67.5% when the newer process is used.

Coating overspray at Pennsylvania House is collected on dry filters that are compressed and stored in 55-gal drums for disposal by landfill. One drum, at a disposal cost of $150/55 gal drum, can hold about 200 compacted filters and solid debris. Waste products generated will include dry and solvent-laden filters and nitrocellulose dust, both loose and trapped in the filters. No liquid waste was generated. Because Pennsylvania House does not separate waste by production lines, no physical data were available for the solid wastestream analysis. However, discussions with Pennsylvania House management and staff consistently indicated that the solid and liquid wastestreams were unaffected by the conversion to the supercritical CO_2 technology.

The chair-finishing process is the same except for the application of the nitrocellulose lacquer finish. Using the old process, two booths were in operation which required cleaning and maintenance. With the UNICARB™ process, only one booth is needed. The transfer efficiency of the modified UNICARB™ spray gun and the air-assisted spray gun are both approximately 50% based on Union Carbide records. However, due to the increased solids content of the UNICARB™ formulation, more solid waste is generated from the overspray by the UNICARB™ process. The 28 dry filters in each of the two spray booths needed for the conventional two-coat finishing process were changed once per week for a total disposal rate of 56 filters/week. The 28 dry filters in the one spray booth required for UNICARB™ finishing are changed twice per week for a total of 56 filters/week. Dry paint and dust from the booths is packed in the disposal drums with the filters, but no increase or decrease in the total volume of these products was noted. Thus, no change was observed by Pennsylvania House in the volume of solid waste generated by converting to the supercritical CO_2 spray process on the chair line.

ECONOMIC ANALYSIS

The objective of the economic analysis was to determine the payback period for the switch to the supercritical CO_2 spray process from the previously used conventional process. The initial investment in capital equipment and installation costs were consider along with the operating costs(materials, waste disposal, labor and utilities). The return-on-investment (ROI) was calculated based on the costs associated with capital expenditures, including equipment and installation, and the return on this investment generated through lower personnel, operating, and materials costs. Details on the ROI calculation are included in the full report.(1)

Implementing the UNICARB™ finishing process on the chair line at Pennsylvania House resulted in substantial annual savings in both utilities and labor as show in Table 3. The annual operating costs were based on the production of 50,000 chairs per year. The UNICARB™ process costs of $ 46,000 include $ 37,000 for the coatings formulation and CO_2 concentrate and $ 9,000 for the CO_2 equipment rental. The conventional formulation costs for the same number of furniture units would be $ 47,000. By converting from a two-coat process to the one-coat process, Pennsylvania House was able to decrease its utility costs by $ 11,000 because there was one less booth to operate. Labor costs were reduced by $ 46,000 because one less finisher and one less sander were needed. Waste handling and disposal costs and finishing line maintenance remained the same for both processes.

Cost savings, realized from a decreased in raw materials costs, were offset by the leasing fees for the CO_2 tank and pump at Pennsylvania House. Additional savings could be realized by decreasing the size of the existing ovens to reflect the change to a one-coat system. Pennyslvania House did not downsize the production ovens to gain gas utility savings.

The annual operating costs of the supercritical CO_2 finish line is approximately $ 58,000 less per year than that of the conventional line. The initial capital investment for the UNICARB™ process was $ 58,000 of which $ 46,000 was for equipment purchase and $ 12,000 for installation of the equipment.

The more detailed economic evaluation found in the full report demonstrates a positive return on investment after the first year with a total payback period with three years if gas utility savings are included, and five years if gas utilities are not included.

CONCLUSIONS

This technology evaluation shows that supercritical CO_2 spray technology has potential as a pollution prevention option in the application of solvent-borne coatings. This technology is not limited to one coating type, but could be used to reduce the solvent level required to spray apply a variety of solvent-borne coatings. The wood furniture facility where this evaluation was conducted maintained product quality with a nitrocellulose lacquer finish and reduced VOC emissions from the coating process. No additional wastes entered the wastestream. Immediate operating savings of $ 58,000/yr were realized. A 100% ROI should be achieved within five years after implementation.

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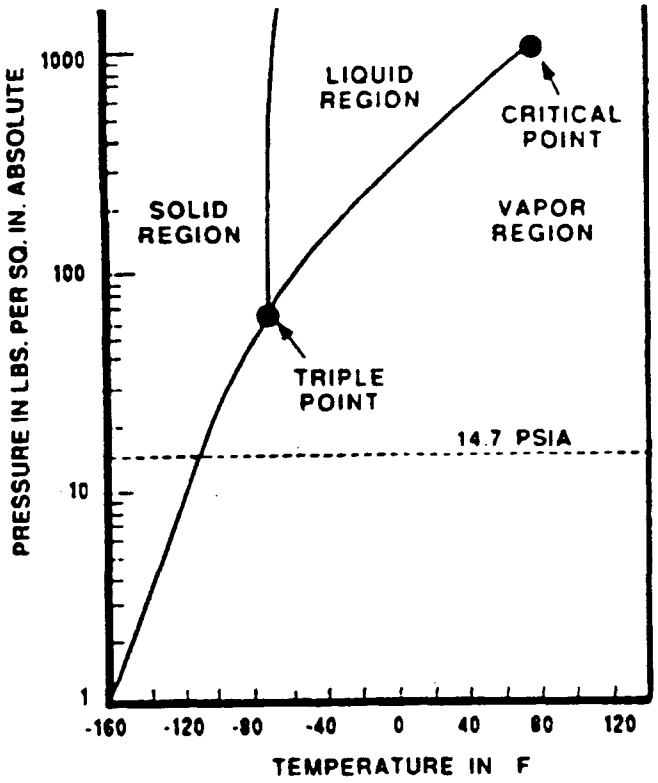

Figure 1. Phase Diagram for carbon dioxide.*

* Nielsen, K.A., Busby, D.C., Glancy, C.W., Hoy, K.L., Kuo,A.C., and Lee, C., "Supercritical Fluid Spray Application Technology: A Pollution Prevention Technology for the Future," Union Carbide Chemicals and Plastics Co., Inc. Presented at the 17th Water-Borne & Higher-Solids Coatings Symposium, February 21-23, 1990, New Orleans, LA.

TABLE 1. Gloss data on sample panels

Finishing Process	Sample Number	Average Gloss Data/Panel	Average Gloss Data/Set
Conventional One Coat	46482-9-1	20.3 ± 4.3	20.3
	46482-9-2	20.4 ± 3.1	
	46482-9-3	20.3 ± 3.1	
Conventional Two Coat	46482-10-1	33.2 ± 1.6	32.3
	46482-10-2	35.0 ± 2.2	
	46482-10-3	28.7 ± 2.8	
UNICARB™	46482-11-1	35.3 ± 3.2	31.5
	46482-11-2	30.5 ± 3.1	
	46482-11-3	28.7 ± 2.9	

TABLE 2. Comparison of volatile solvent content of conventional and UNICARB™ coating formulations as percent weight.

Materials Description	HAP (Y/N)	Conventional (% by weight)	UNICARB™ (% by weight)
MEK-heptanone	No		37.25
methoxypropylacetate	No	7.36	
xylene	Yes	16.80	
isopropanol	No	11.20	6.55
toluene	Yes	10.39	
N-butyl acetate	No	11.83	
isobutyl acetate	No	6.89	
2-butoxyethanol	Yes	3.27	9.67
MIBK	Yes	5.32	
isopropyl acetate	No	1.46	
Other		2.37	5.87
Total VOC (% by weight)		**76.88**	**59.34**

Table 3. Summary of Annual Operating Cost Comparing Conventional Finish Line with Supercritical CO_2 Finish Line

Item	Conventional ($/year)	Supercritical CO_2 ($/year)
Coating Materials	47,000	37,000
CO_2 Storage Equipment	---	9,000
Spray Booth #2		
Finish Labor	23,000	---
Sanding Labor	23,000	---
Electricity	11,000	---
TOTAL	104,000	46,000